PROJECT REPORT FOR

Foundations of Artificial Intelligence

STUDYQUEST AI: A GAME-BASED STUDY PLANNER

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report ": A GAME-BASED STUDY PLANNER" is the Bonafide work of "THENDRAL V (240701567) AND VARNIKA V(240701578)**" who carried out the project work under my supervision.**

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ABSTRACT

The increasing academic workload and lack of effective study planning can make it difficult for students to manage their time, understand complex subjects, and stay motivated during preparation. This project, StudyQuest AI: A Game-Based Study Planner, presents an interactive web application that combines study planning with gamification to make learning more engaging and organized. The system provides a dashboard to track experience points (XP), completed quests, and study streaks. Users can generate a study quest route by selecting a subject, available study time, exam date, and planning strategy. The application divides the learning process into structured activities, including concept learning, practice sessions, quizzes, and boss battles.

The system incorporates game-based elements such as levels, rewards, achievement badges, progress tracking, countdown timers, and interactive quizzes to encourage consistent study habits. Students can complete learning objectives to finish quests and earn experience points. The boss battle feature provides a question-based challenge in which correct answers reward the user with additional XP, while incorrect answers reduce the opponent's health points. The application is developed using HTML, CSS, and JavaScript, with a responsive user interface and rule-based interactions. The current implementation uses predefined quest data and fixed quest-generation logic rather than a trained machine learning model or an external AI service. StudyQuest AI demonstrates how gamification and structured study activities can support student motivation, task completion, and time management. Future enhancements may include personalized recommendations, adaptive scheduling, performance analytics, and integration with machine learning techniques to provide a more intelligent and personalized learning experience.

CHAPTER 1

INTRODUCTION

Students often face difficulties in managing their academic workload, organizing study schedules, and maintaining motivation throughout their learning process. Traditional study-planning methods mainly depend on manually created timetables and task lists, which may not provide enough engagement or encouragement to maintain consistent study habits. This project, StudyQuest AI: A Game-Based Study Planner, addresses these challenges by combining study planning with gamification to create a more interactive and engaging learning experience.

The proposed system is a web-based application that allows students to organize their study activities through quests and track their progress using a dashboard. Students can select a subject, specify their available study time, enter an exam date, and choose a planning strategy to generate a structured quest route. The route consists of learning concepts, practicing problems, attempting quizzes, and completing a final boss battle. Each activity is designed to divide the learning process into manageable tasks and encourage students to progress step by step.

StudyQuest AI incorporates game-based features such as experience points (XP), levels, achievement badges, study streaks, progress indicators, countdown timers, and interactive challenges. Completing quests increases the student's experience points and completed-quest count, while the boss battle provides an additional challenge through question-based interaction and rewards. These features aim to encourage regular participation, improve task completion, and make study sessions more enjoyable.

The application is implemented using HTML, CSS, and JavaScript, providing an interactive interface with dashboard navigation, quest management, timer controls, quiz interaction, and reward tracking. The current version uses predefined quest information and rule-based functionality to generate study routes and manage game progress; it does not implement a trained machine learning model or connect to an external AI service. Therefore, the project demonstrates the practical application of gamification and structured task planning in an educational environment. Future enhancements may include adaptive study schedules, personalized recommendations based on student performance, persistent progress storage, and machine learning-based analysis to support more personalized learning experiences.

CHAPTER 2 LITERATURE

REVIEW

Author & Year	Methodology	Drawback	Improvement in StudyQuest AI
Deterding et al. (2011)	Introduced the concept of gamification by applying game-design elements in non-game contexts.	Provides a conceptual foundation but does not present a complete study-planning application.	Applies gamification through quests, XP points, levels, badges, and progress tracking.
Deci & Ryan (2000)	Examined self-determination theory, focusing on autonomy, competence, and relatedness as factors influencing motivation.	The theory does not provide a specific software implementation for academic planning.	Uses selectable study preferences, achievable quests, and progress indicators to encourage student motivation.
Kapp (2012)	Explained the application of game-based learning, game mechanics, and instructional design in education and training.	Provides broad instructional guidance rather than a specific personalized study-planning system.	Combines structured learning tasks with interactive challenges and rewards in a single web application
Hamr, Koivisto & Sarsa (2014)	Reviewed empirical studies investigating the effects of gamification across different contexts.	Reported that gamification outcomes vary depending on the context and implementation.	Integrates multiple game elements, including quest completion, experience points, streaks, and achievements, to support study engagement.

Sailer et al. (2017)	Conducted an experimental study examining how specific game-design elements influence psychological need satisfaction and motivation.	Examines selected game elements in an experimental setting rather than implementing a complete study planner.	Incorporates progress feedback, achievement rewards, and challenge-based activities into the study workflow.
Dichev & Dicheva (2017)	Conducted a critical review of gamification in education, examining its effectiveness and the limitations of existing evidence.	Highlighted uncertainty regarding the long-term educational benefits of gamification.	Provides a practical application for organizing study tasks while making progress visible through interactive dashboard features.
Dunlosky et al. (2013)	Reviewed learning techniques such as practice testing and distributed practice to evaluate their effectiveness for student learning.	Focuses on learning techniques rather than an integrated, gamified study-planning application.	Includes quiz-based challenges and structured study sessions that can support active practice and organized learning.
Werbach & Hunter (2012)	Explained how game elements, mechanics, and dynamics can be used to design gamified systems	Provides a general framework and does not specifically address automated academic quest generation.	Uses quests, XP, achievement badges, and boss battles to structure the student's learning journey.
Russell & Norvig (2021)	Presented foundational artificial intelligence concepts, including intelligent agents, problem-solving, search strategies, and decision-making.	Provides general AI theory rather than a ready-made educational gamification platform.	Establishes a foundation for future extensions involving intelligent study recommendations and adaptive planning.

Table 2.1 literature survey

The literature review examines nine studies covering gamification, student motivation, educational learning techniques, and artificial intelligence foundations to understand existing approaches and identify opportunities for improvement in StudyQuest AI. Table 2.1 summarizes the methodology of each study, its principal limitations, and how the proposed system applies the relevant concepts.

A common theme across these studies is that theoretical frameworks explain motivation and learning, while many gamification approaches focus on individual game elements rather than integrating them into a structured study-planning application. Research on self-determination theory highlights the importance of motivation and competence, while studies on effective learning techniques emphasize the value of practice testing and organized learning activities. However, these concepts alone do not provide students with a single interactive platform for organizing study tasks and tracking their progress.

StudyQuest AI addresses this gap by combining study planning and gamification within one web application. It provides a quest-based dashboard, subject selection, study-time and exam-date inputs, structured learning activities, countdown timers, quiz challenges, experience points, and achievement badges. Unlike a conventional timetable or task list, the system presents study activities as progressive challenges that students can complete and track. The current implementation uses predefined quest data and rule-based interactions, providing a foundation for future enhancements such as adaptive scheduling, personalized recommendations, and performance-based study planning.

CHAPTER 3

PROPOSED SYSTEM

StudyQuest AI is designed to provide an interactive and engaging study-planning experience by combining structured learning activities with game-based elements. Instead of relying entirely on traditional timetables and task lists, the system organizes academic activities into quests that students can complete while tracking their progress and achievements.

The application provides a dashboard through which students can view their experience points (XP), completed quests, study streaks, and active study activities. Students can navigate to the Quest Map and select a subject, available study time, exam date, and preferred planning strategy. Based on these inputs, the system generates a predefined quest route consisting of concept learning, practice activities, an AI quiz, and a boss battle. These activities help divide the study process into manageable stages.

The Quest Detail module allows students to view the description, estimated duration, and XP reward associated with a selected quest. A countdown timer supports study sessions through start, pause, and reset controls. Students must complete three learning objectives—reading the concept, practicing examples, and completing the quiz—before the Complete Quest button becomes available. Upon completion, the application updates the experience points and completed-quest count and displays a confirmation message.

The Missions module introduces an interactive quiz challenge in which students answer questions to defeat a virtual opponent. Correct answers can defeat the boss and award additional XP, while incorrect answers reduce the opponent's health points. The Rewards module displays achievement badges such as Quest Master, Streak Warrior, Quiz Slayer, and Boss Hunter, encouraging students to work toward learning milestones.

The system is implemented using HTML, CSS, and JavaScript. HTML structures the application pages, CSS provides the visual layout and responsive styling, and JavaScript manages navigation, quest generation, timer operations, quiz interactions, progress updates, and reward-related feedback. The current version uses predefined quest data and rule-based logic; it does not include a trained machine learning model or an external AI service. The proposed system therefore demonstrates how game-based interaction and structured study activities can be combined to support student engagement and study organization, while providing a foundation for future adaptive and personalized planning features.

3.1 ARCHITECTURAL DIAGRAM

The architecture of the proposed system can be represented as follows:

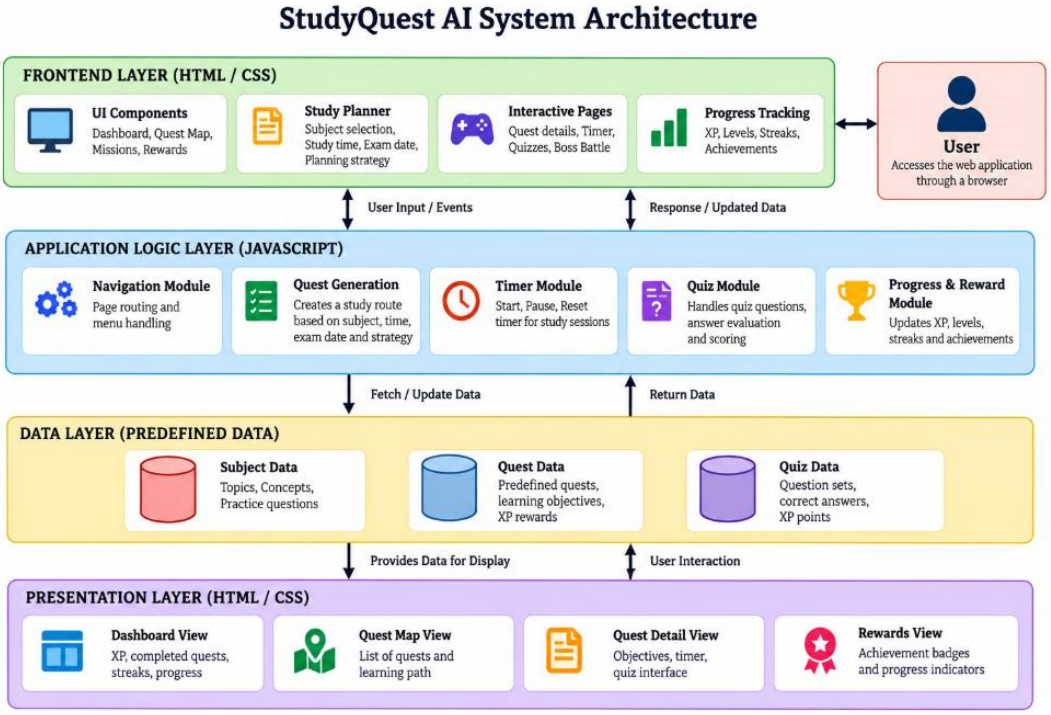


figure 3.1 Architecture diagram

The user interacts with StudyQuest AI through the web application, where the Dashboard displays experience points, completed quests, study streaks, and active study activities. The student can navigate to the Quest Map, select a subject, specify available study time and an exam date, and choose a planning strategy. The JavaScript application processes these inputs and generates a predefined quest route consisting of concept learning, practice activities, an AI quiz, and a boss battle.

The quest details, study timer, task completion, quiz interactions, and reward calculations are managed by the application's JavaScript logic. Students can start, pause, or reset the timer and complete the required learning objectives to finish a quest. Upon completion, the application updates the experience points and completed-quest count and displays feedback. The Missions module provides an interactive boss battle, while the Rewards module displays achievement badges and progress information. All components work together through the HTML, CSS, and JavaScript implementation to provide an interactive, game-based study-planning experience. The current version uses predefined quest data and does not include a backend server or database for persistent storage.

CHAPTER 4

MODULES DESCRIPTION

The StudyQuest AI system is divided into five main modules to provide a structured and interactive study-planning experience. Each module performs a specific function, including dashboard management, quest generation, study-session tracking, interactive quiz challenges, and reward management. Together, these modules help students organize their learning activities and monitor their progress through game-based features.

4.1 MODULE 1 – DASHBOARD AND NAVIGATION

This module provides the main interface through which students access the different features of StudyQuest AI. The dashboard displays the student's experience points (XP), completed quests, current study streak, and active quests. The navigation system allows users to move between the Dashboard, Quest Map, Missions, and Rewards pages. This module provides a centralized view of study progress and makes the application's features easily accessible.

4.2 MODULE 2 – STUDY PLANNING AND QUEST GENERATION

This module enables students to generate a structured study route by selecting a subject, available study time, exam date, and planning strategy. Based on the selected inputs, the application generates a predefined sequence of activities: Concept Quest, Practice Battle, AI Quiz, and Boss Battle. Each activity represents a stage in the learning process. The current implementation uses predefined quest-generation logic rather than a machine learning model or an adaptive AI algorithm.

4.3 MODULE 3 – QUEST MANAGEMENT AND STUDY TIMER

This module manages individual study quests and helps students monitor their study sessions. Each quest displays its title, description, estimated duration, and XP reward. Students are provided with a countdown timer that supports start, pause, and reset operations. The module also includes three learning objectives: reading the concept, practicing examples, and completing the quiz. The Complete Quest button is enabled only after all three objectives have been selected. Once the quest is completed, the application updates the XP total and completed-quest count and displays a confirmation message.

4.4 MODULE 4 – MISSIONS AND BOSS BATTLE

This module introduces an interactive question-based challenge to make the learning process more engaging. The Missions page presents a Python function question with multiple answer choices and a virtual opponent called the Code Goblin. The opponent begins with 100 health points. A correct answer defeats the opponent and awards additional XP, provided the boss has not already been defeated. An incorrect answer reduces the opponent's health by 20 points and displays feedback. This module combines quiz interaction with game-based challenges to encourage active participation.

4.5 MODULE 5 – REWARDS AND PROGRESS TRACKING

This module displays the student's progress and achievement badges. The application tracks experience points, completed quests, and progress toward the next XP milestone. It also provides achievement cards such as Quest Master, Streak Warrior, Quiz Slayer, and Boss Hunter, each representing a specific milestone. These reward elements are designed to encourage students to complete study activities and maintain consistent participation. Progress values are managed using JavaScript during application use; the current version does not include a database for permanent progress storage.

CHAPTER 5

IMPLEMENTATION AND RESULTS

The StudyQuest AI application was implemented as an interactive web-based study planner that combines structured study activities with game-based features. The system allows students to generate quest routes, complete learning objectives, use a study timer, participate in quiz challenges, and track their experience points and achievements. The application provides visual feedback whenever students complete quests or interact with the boss battle.

5.1 EXPERIMENTAL SETUP

The StudyQuest AI system is implemented using HTML, CSS, and JavaScript. HTML defines the structure of the application pages, including the Dashboard, Quest Map, Missions, and Rewards. CSS provides the visual design, layout, colors, and responsive styling. JavaScript manages page navigation, predefined quest data, quest-route generation, countdown timer operations, learning-objective validation, quiz interactions, experience-point calculations, and progress updates.

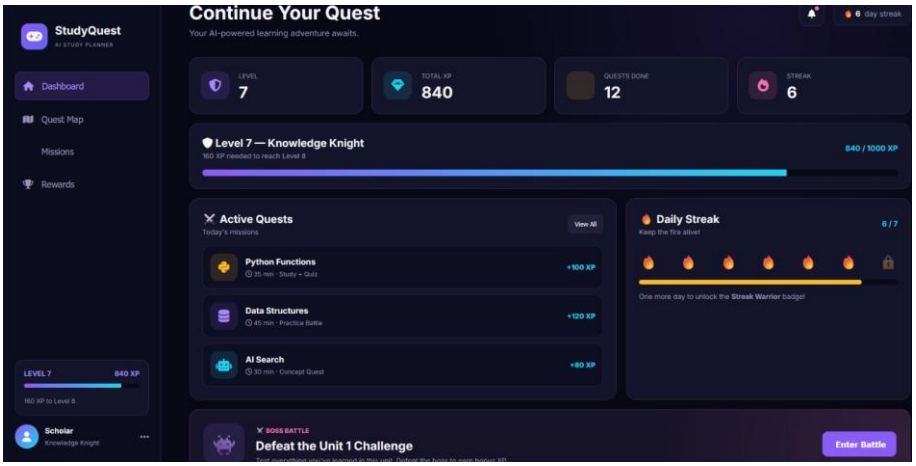
The application was tested through its interactive features to verify that students can navigate between pages, generate a quest route by entering study preferences, open individual quests, operate the study timer, and complete the required learning objectives. The Missions module provides a Python function quiz and a boss battle in which correct and incorrect answers affect the opponent's health points. The Rewards module displays achievement badges, while the dashboard reflects updates to experience points and completed quests.

The current implementation uses predefined quest information and rule-based JavaScript logic. It does not use a trained machine learning model, an external AI service, or a backend database for persistent storage. The application demonstrates how gamification features can be integrated into a web-based study planner to encourage student engagement and organize learning activities.

5.1 RESULTS

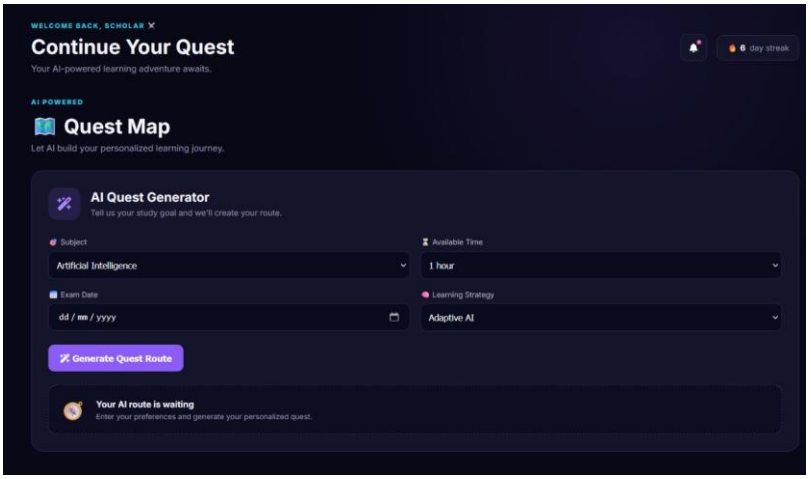
5.1.1 StudyQuest AI Dashboard

The dashboard displays the student's current experience points (XP), completed quests, study streak, and active study activities. It provides a centralized view of the student's progress and allows navigation to the Quest Map, Missions, and Rewards modules.



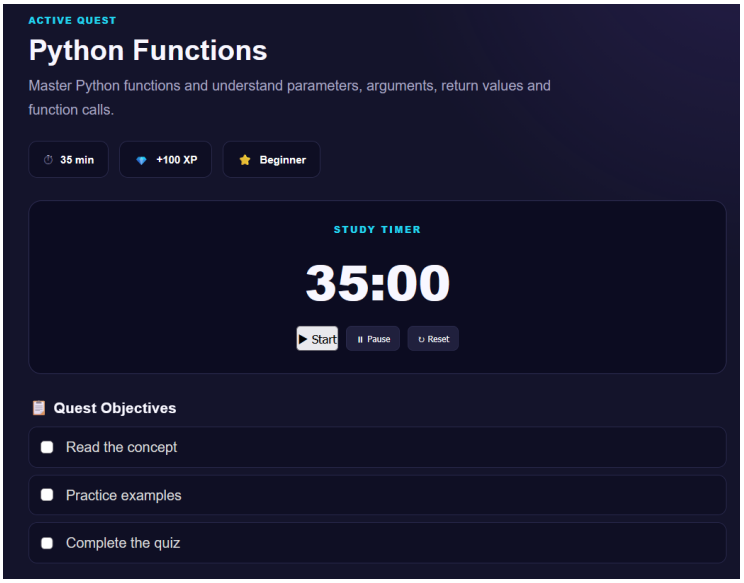
5.1.2 Quest Map and Study Route Generation

The Quest Map allows students to select a subject, enter their available study time and exam date, and choose a planning strategy. Based on these inputs, the application generates a predefined study route consisting of concept learning, practice activities, an AI quiz, and a boss battle.



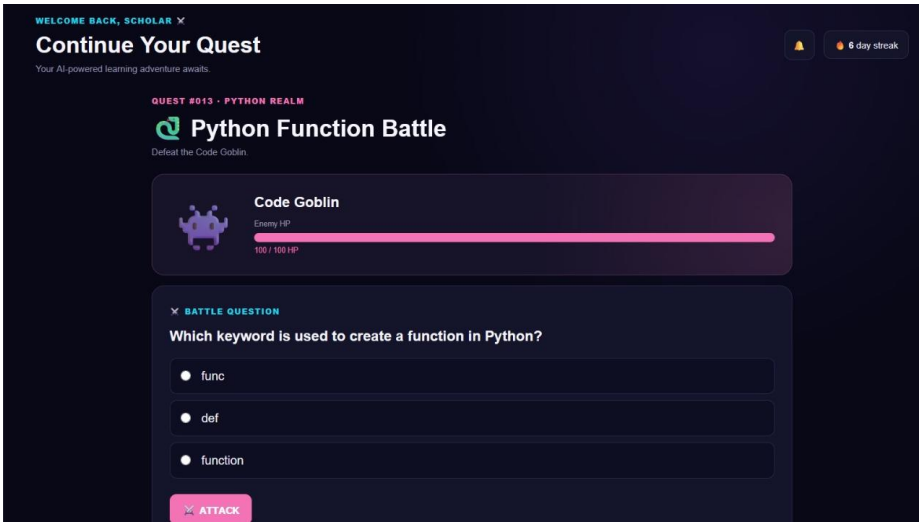
5.1.3 Quest Details and Study Timer

The Quest Detail page displays the selected quest's title, description, estimated duration, and XP reward. The countdown timer supports start, pause, and reset operations. Students must select all three learning objectives before completing the quest, helping them follow a structured study process.



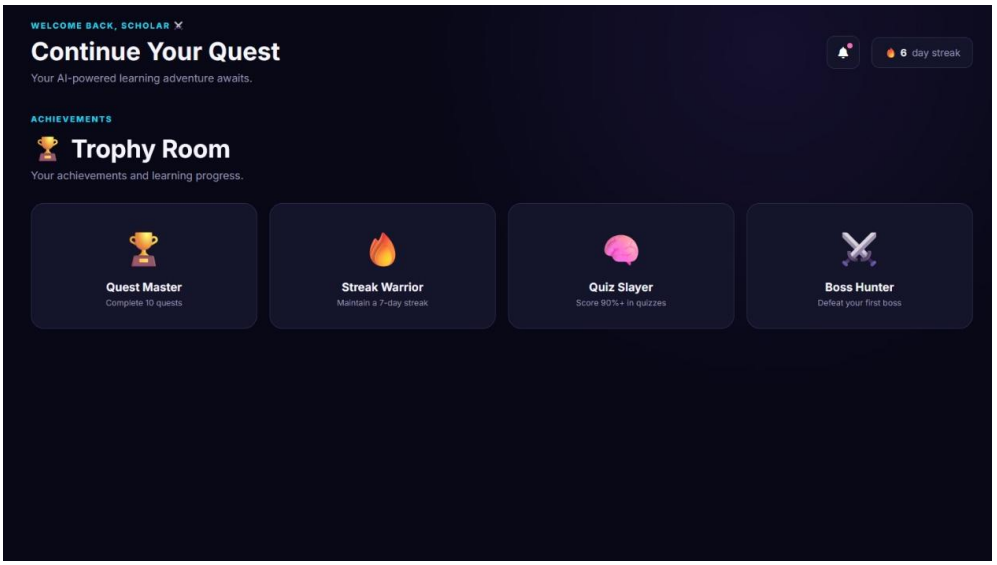
5.1.4 Missions and Boss Battle

The Missions module presents an interactive Python quiz and a virtual opponent named Code Goblin. Correct answers defeat the opponent and award additional XP if it has not already been defeated, while incorrect answers reduce its health points. This module demonstrates the integration of quiz-based learning and game mechanics.



5.1.5 Rewards and Progress Tracking

The Rewards module displays achievement badges such as Quest Master, Streak Warrior, Quiz Slayer, and Boss Hunter. The application also updates experience points and completed-quest counts when quests are completed, providing visual feedback on the student's progress.



The StudyQuest AI application was tested to evaluate its main functionalities, including quest generation, study-session management, quiz interaction, experience-point updates, and achievement display. The results were observed through the application's dashboard and individual modules to verify that the different components work together as intended.

The quest-generation module successfully creates a predefined study route based on the selected subject, available study time, exam date, and planning strategy. The Quest Detail module provides a countdown timer with start, pause, and reset controls. It also verifies that all three learning objectives are selected before enabling the Complete Quest button. After a quest is completed, the application updates the experience points and completed-quest count and displays a confirmation message.

The Missions module provides an interactive Python quiz and a boss battle. A correct answer defeats the opponent and awards additional experience points if the boss has not already been defeated, while an incorrect answer reduces the opponent's health points. The Rewards module displays achievement badges, and the dashboard presents updated progress information. These features demonstrate the integration of game-based challenges and progress tracking within the study-planning application.

Overall, the implementation demonstrates the practical use of web technologies and rule-based interactions to create a gamified study planner. The application provides students with a structured way to organize study activities, monitor quest completion, and engage with learning challenges. However, formal measurements of improvements in academic performance, long-term motivation, or study efficiency have not yet been established. Future testing with students and the collection of usage data could help evaluate the educational effectiveness of StudyQuest AI more thoroughly.

CHAPTER 6

CONCLUSION AND FUTUREWORK

The project presents StudyQuest AI: A Game-Based Study Planner, a web-based application designed to make study planning more interactive and engaging through gamification. By combining structured study quests, countdown timers, interactive quizzes, experience points, achievement badges, and progress tracking, the system provides students with an organized approach to managing their learning activities. The application uses HTML, CSS, and JavaScript to implement its interface and core functionality, with predefined quest data and rule-based interactions.

Testing of the implemented features demonstrated that students can generate study quest routes based on their selected preferences, complete learning objectives, manage study sessions using the timer, and participate in quiz-based boss battles. The application updates experience points and completed-quest counts when quests are completed and provides visual feedback through its dashboard and rewards interface. These features demonstrate the practical integration of game-based elements into a study-planning application. However, the current implementation does not include a trained machine learning model or an adaptive AI engine, and its effectiveness in improving academic performance has not yet been formally evaluated.

As future work, StudyQuest AI can be enhanced with personalized study recommendations based on student performance, adaptive quest generation according to topic difficulty, and automated scheduling based on exam dates and available study hours. Database integration could enable persistent storage of student progress, completed quests, and achievements across sessions. Additional features could include a larger question bank, subject-specific quizzes, detailed performance analytics, and reminders for upcoming study sessions. Further evaluation with students could help measure the application's impact on study consistency, motivation, and time management. These improvements could transform StudyQuest AI into a more personalized and intelligent learning platform.

CHAPTER 7

REFERENCES

1. Deterding, S., Dixon, D., Khaled, R. and Nacke, L. (2011) 'From Game Design Elements to Gamefulness: Defining "Gamification"', Proceedings of the 15th International Academic MindTrek Conference, pp. 9–15. Available at: https://doi.org/10.1145/2181037.2181040

2. Deci, E.L. and Ryan, R.M. (2000) 'The "What" and "Why" of Goal Pursuits: Human Needs and the Self-Determination of Behavior', Psychological Inquiry, 11(4), pp. 227–268. Available at: https://doi.org/10.1207/S15327965PLI1104_01

3. Dunlosky, J., Rawson, K.A., Marsh, E.J., Nathan, M.J. and Willingham, D.T. (2013) 'Improving Students' Learning With Effective Learning Techniques: Promising Directions From Cognitive and Educational Psychology', Psychological Science in the Public Interest, 14(1), pp. 4–58. Available at: https://doi.org/10.1111/j.1539-6053.2013.01038.x

4. Hamari, J., Koivisto, J. and Sarsa, H. (2014) 'Does Gamification Work?—A Literature Review of Empirical Studies on Gamification', Proceedings of the 47th Hawaii International Conference on System Sciences, pp. 3025–3034. Available at: https://doi.org/10.1109/HICSS.2014.377

5. Kapp, K.M. (2012) The Gamification of Learning and Instruction: Game-Based Methods and Strategies for Training and Education. San Francisco: Pfeiffer.

6. Dichev, C. and Dicheva, D. (2017) 'Gamifying Education: What Is Known, What Is Believed and What Remains Uncertain: A Critical Review', International Journal of Educational Technology in Higher Education, 14, Article 9. Available at: https://doi.org/10.1186/s41239-017-0042-5

7. Sailer, M., Hense, J.U., Mayr, S.K. and Mandl, H. (2017) 'How Gamification Motivates: An Experimental Study of the Effects of Specific Game Design Elements on Psychological Need Satisfaction', Computers in Human Behavior, 69, pp. 371–380. Available at: https://doi.org/10.1016/j.chb.2016.12.033

8. Werbach, K. and Hunter, D. (2012) For the Win: How Game Thinking Can Revolutionize Your Business. Philadelphia: Wharton Digital Press.

9. Russell, S. and Norvig, P. (2021) Artificial Intelligence: A Modern Approach. 4th edn. Harlow: Pearson.

10. Mozilla Developer Network (MDN) (n.d.) HTML: HyperText Markup Language. Available at: https://developer.mozilla.org/en-US/docs/Web/HTML (Accessed: 21 October 2026).

11. Mozilla Developer Network (MDN) (n.d.) CSS: Cascading Style Sheets. Available at: https://developer.mozilla.org/en-US/docs/Web/CSS (Accessed: 21 October 2026).

12. Mozilla Developer Network (MDN) (n.d.) JavaScript. Available at: https://developer.mozilla.org/en-US/docs/Web/JavaScript (Accessed: 21 October 2026).